

One dataset for the whole archive, and what imputation can do for it

Every station, both instrument eras, 26 July 2018 to 13 August 2026

Study report · `studies/unified_dataset/`

13 August 2026

Abstract

Every earlier study in this project began by reading the raw archive and ended by discarding most of it. This one builds the dataset those studies should have started from — all three legacy stations, both column eras, cleaned, compensated and carrying provenance on every row — and then asks what can be done about the holes.

The archive is emptier than its span suggests. Eight years of calendar time contain **2061 days** of inclinometer readings at the target station, 70 % of the span; the other two stations hold 1119 and 984 days and both stopped recording years before the February 2025 instrument replacement. The network narrowed from three locations to one in stages, not at a single moment.

The missingness is the worst kind available. On every gap, every channel is missing at once: air temperature, humidity, battery and inclinometer fail together on **100 %** of the missing hours. This is outage-driven missingness, and it has a consequence that governs the whole study — no method conditioned on a covariate can ever fill a real gap here, because the covariates are gone with the target.

The benchmark cannot see that, and the study reports the discrepancy. Injected gaps have their drivers intact, so a driver-based filler scores best at almost every length and would be useless in production. This is the exchangeability assumption behind any conformal guarantee, failing concretely and measurably.

The two eras are two instruments, and a constant does not reconcile them. The step at the changeover is -132.5 mdeg once the season is removed, but the diurnal amplitude rises from 13.4 to 17.9 mdeg across it — a **ratio of 1.34**. The instruments differ in gain as well as in zero, so the joined series is offered as a clearly named derived column and never as the default.

The record does not have to stay in pieces. Without filling anything, the longest strictly continuous stretch in eight years is **108 days**. Imputation raises that to **988.9 days** — **2.71 annual cycles** — at 14.5 % invented, or to **617.2 days** at 3.2 % if the stretch is required to stay within one instrument era. Both carry a calibrated interval on every invented value, and past a 15 mdeg tolerance no further record can be bought at any price.

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1 What this study builds, and why

1.1 Two deliverables

The work is split into two notebooks, deliberately.

`build_unified_dataset.py` is the **ingest**, and it is the only place in the project that reads the raw archive. It is meant to be re-run whenever the archive grows: it discovers the extent itself, so no date has to be edited. It writes one table per station to `data/interim/unified/`, plus a manifest recording how the build was made.

`imputation_study.py` is this report. It consumes that output and never touches an `.adc` file.

The separation is not tidiness. The ingest writes only measurements, and every row it writes carries `inc_source = observed`. Imputation happens on the far side of that boundary, so a filled value can never be mistaken for a reading by a notebook three steps downstream.

1.2 Three rules the dataset obeys

The rules, and where they come from

The two eras are two instruments. The package installed on 2025-02-21 replaced the legacy network rather than augmenting it, and its inclinometer shares no baseline with legacy block b2 (`docs/raw-data-format.md` Section 3.3). The primary `inc` and `inc_comp` columns keep the baselines apart; `era` and `instrument` say which is which.

The absolute level means nothing. Only changes carry structural information (Section 7.1 of the same document). Every level here is referred to an arbitrary anchor.

A filled value is not a measurement. Provenance is carried on every row, through every product, to the end.

1.3 What the dataset contains

One CSV per station on the hourly grid, spanning the whole archive.

Column	Meaning
batt, tair, rh	Supply voltage, air temperature, relative humidity
inc	Inclinometer, millidegrees about the instrument’s calibrated zero. Its absolute level carries no structural information
sr, twall	Solar radiation and wall temperature. Current era, target station only
inc_comp	Compensated at the documented coefficient, anchored <i>within each era separately</i>
inc_comp_joined	The same with the estimated instrument offset removed, so the eras form one series. Target station only. Carries an estimated constant
era, instrument	Which era, and which physical instrument
inc_source	observed or missing . The ingest fills nothing

The compensation uses the documented coefficient of 5 mdeg per °C. That value has never been calibrated for these instruments, and the legacy study measured its influence on a headline error at −58 % to +183 %. It is used here for continuity with every earlier study, and the reader should treat absolute levels accordingly.

2 What is actually there

2.1 Three stations, three different records

Station	Eras	Last reading	Span [d]	Observed [d]	of archive
st01	legacy	2022-06-01	1406.3	1118.7	38.0 %
st02	legacy+current	2026-08-13	2940.0	2061.0	70.1 %
st03	legacy	2023-06-24	1794.6	984.1	33.5 %

The network failed in stages, not at the changeover

The data-quality report records that the current-era package “replaced” the legacy network, which is true of the file format and misleading about the instrumentation. **Station 01 stopped recording in June 2022 and station 03 in June 2023** — two years and eight months before the replacement respectively.

For most of the archive there was never a three-station network to compare across. Any cross-station test — including the one most likely to settle the sign contradiction documented in the two prediction studies — is confined to the period before June 2022.

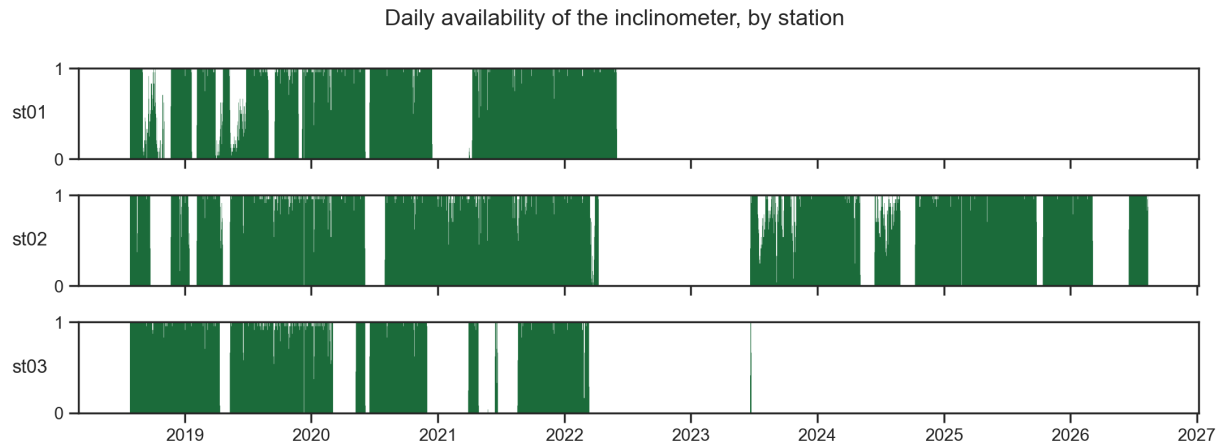


Figure 1: Daily availability of the inclinometer at each station. The staggered right-hand edges are the point: the three records end three years apart.

2.2 The target station, era by era

Era	From	To	Span [d]	Cycles	inc	twall
legacy	2018-07-26	2025-02-20	2401.0	6.57	68.5 %	—
current	2025-02-21	2026-08-13	538.0	1.47	77.2 %	36.8 %

The current era is the better-instrumented and the better-covered of the two, and it is short. Wall temperature is present on barely a third of it, because the probe-failure sentinel and the acquisition outages interleave.

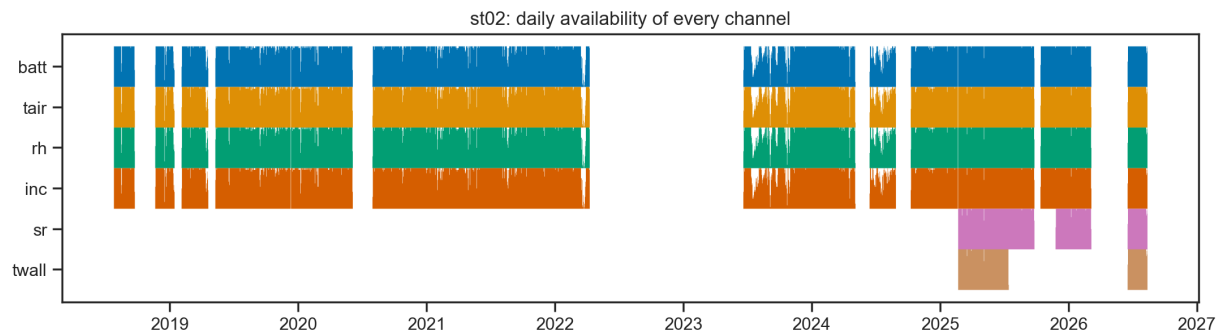


Figure 2: Daily availability of every channel at the target station. Solar radiation and wall temperature begin only at the changeover; wall temperature is then interrupted repeatedly by its own failure sentinel.

2.3 The calendar

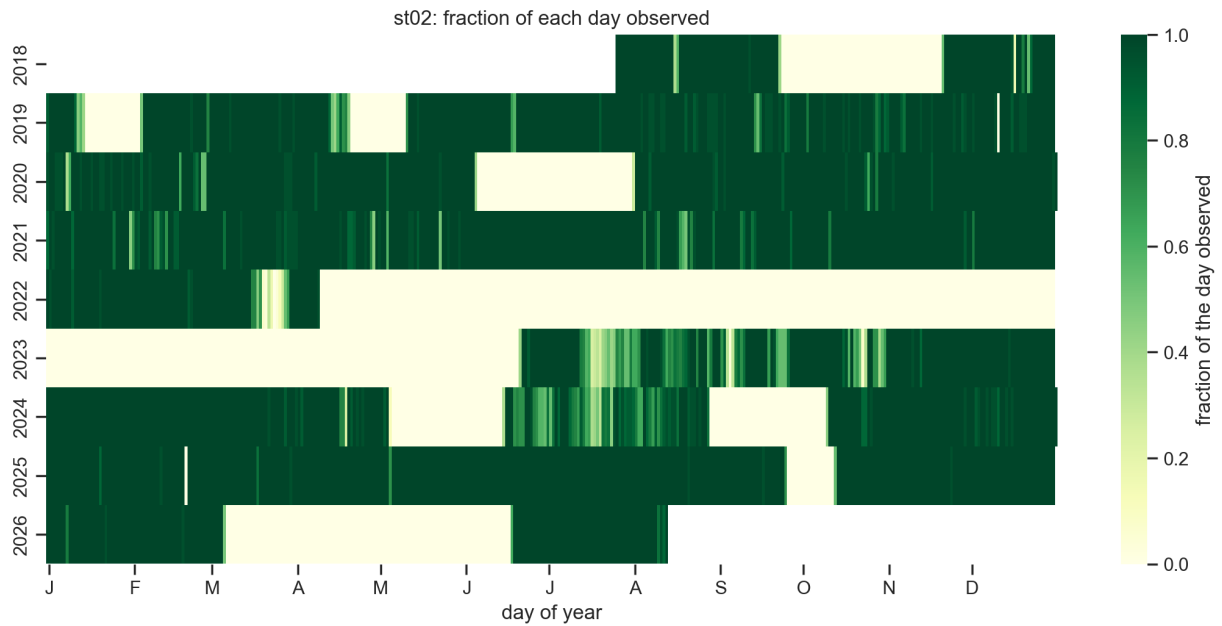


Figure 3: Fraction of each day observed, by year and day of year. Outages appear as horizontal bands and are not confined to any season.

3 The instrument changeover

The two eras are adjacent — the legacy record ends on 2025-02-19 and the current one begins on 2025-02-21, an interruption of about a day — so a level comparison across the boundary is possible. It is also misleading if taken naively, because the two comparison windows sit at different points of the annual cycle.

Quantity	Value
Interruption	1.04 d
Raw step	−127.67 mdeg
Attributable to the season	+4.81 mdeg
Attributed to the instrument	−132.48 mdeg
Diurnal amplitude, legacy side	13.38 mdeg
Diurnal amplitude, current side	17.91 mdeg
Gain ratio	1.34

A constant offset does not reconcile the two instruments

The offset is estimable and the estimate is reported. The diagnostic beside it is the one that matters: the diurnal amplitude — the signal’s response to the daily thermal cycle, which is a property of the wall and not of the logger — rises by **34 %** across the changeover.

The wall did not become a third more responsive to sunlight overnight. Either the new instrument has a different gain, or it is mounted where the thermal response is larger, or both. A single additive constant cannot absorb a multiplicative difference, so `inc_comp_joined` reconciles the *levels* of the two eras and does not reconcile their *scales*.

This is why the joined series is a derived column and not the default, and why any analysis spanning the changeover should either work in the difference domain or report results per era.

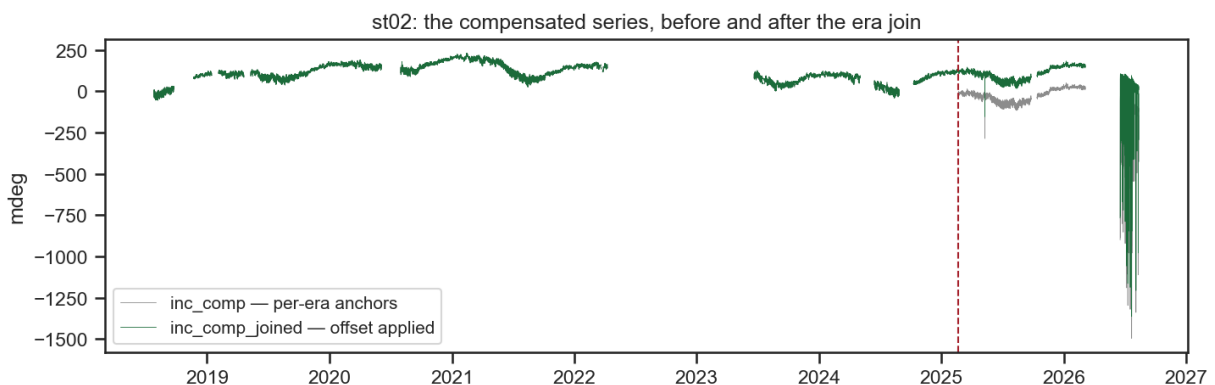


Figure 4: The compensated series before and after the join. The offset removes the level discontinuity; it cannot remove the amplitude difference.

4 What is missing, and why

4.1 The anatomy of the gaps

Band	Gaps	Hours lost	Share	Median
≤ 3 h	426	619	2.9 %	1 h
3–6 h	60	284	1.3 %	4 h
6–24 h	72	906	4.3 %	11 h
1–3 d	2	65	0.3 %	33 h
3–7 d	0	0	0.0 %	—
7–30 d	3	1 415	6.7 %	491 h
> 30 d	6	17 809	84.4 %	1 396 h

569 gaps hold 21 098 missing hours, and the distribution is extremely uneven. **The 560 gaps of three days or less account for 8.9 % of the loss; six gaps longer than thirty days account for 84.4 %.** Almost everything that is missing is missing in a handful of long outages.

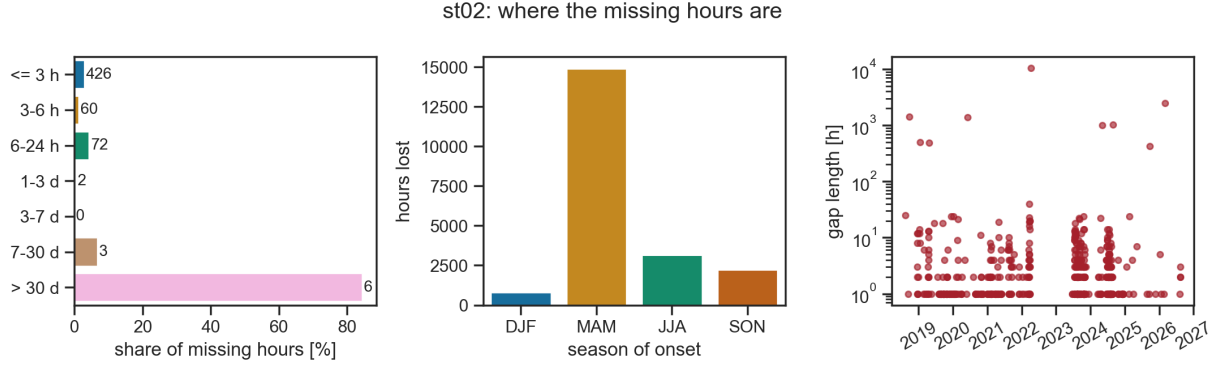


Figure 5: Left: share of missing hours by length band, annotated with the number of gaps. Centre: hours lost by season of onset. Right: every gap, by date and length, on a logarithmic scale.

4.2 The nine that matter

From	To	Days	Season
2022-04-09	2023-06-21	437.6	MAM
2026-03-06	2026-06-18	103.9	MAM
2018-09-23	2018-11-21	59.0	SON
2020-06-04	2020-07-31	57.3	JJA
2024-08-27	2024-10-09	42.7	JJA
2024-05-04	2024-06-14	41.5	MAM
2019-01-14	2019-02-04	20.8	DJF
2019-04-20	2019-05-11	20.5	MAM
2025-09-25	2025-10-13	17.8	SON

These nine cut the eight-year archive into ten pieces. Their onsets are spread across all four seasons, so there is no seasonal maintenance pattern to exploit.

4.3 Why the record is missing

Every channel fails at once

Of the hours in which the inclinometer is missing, the fraction in which each other channel is *also* missing:

batt	100.0 %
tair	100.0 %
rh	100.0 %
sr	100.0 %
twall	100.0 %

There is no covariate that is ever observed while the target is absent. The missingness is not random, and it is not conditional on something the record still contains — it is an **outage**, and an outage takes the whole station with it.

The consequence is hard and it governs Section 5: no method conditioned on a covariate can fill a real gap in this record. Not because such methods are poor, but because their inputs are missing exactly when they would be needed.

This also settles a question the framework in the companion manuscript raises but cannot answer from the extended-sensor era alone. Its regressor-diagnosis stage applies an availability filter — does a candidate exist during the intervals that must be reconstructed? — and warns that

on-structure channels sharing the station’s power and communications will fail it by construction. Here that filter does not merely reduce the candidate pool. **It empties it.**

5 What can be filled, and how well

Six fillers of increasing ambition, measured against gaps of known length injected where the record is complete. No filler sees the values it is asked to reproduce. The range matters because no single method is right at every length: interpolation is unbeatable across an hour and meaningless across a month, while a seasonal model knows nothing useful about an hour and is most of what remains after a week.

Filler	What it knows
interpolate	The two flanking observations, and nothing else
seasonal_naive	That there is a daily cycle
kalman	A local linear trend plus a daily seasonal, in state-space form. The only filler with a native notion of its own uncertainty
harmonic	The annual and diurnal cycles, fitted by least squares. Its error does not grow with gap length
drivers	A ridge regression on air temperature and humidity
neuralprophet	Trend and seasonality from the project’s own decomposition

5.1 Error against gap length

Gap	interp.	seas. naive	kalman	harmonic	drivers
1 h	1.22	4.58	1.04	3.84	1.83
3 h	1.61	5.67	1.75	3.73	<i>1.14</i>
6 h	2.79	4.52	2.99	4.00	<i>2.48</i>
12 h	8.75	9.04	8.55	9.98	<i>7.25</i>
24 h	9.83	5.12	5.71	6.13	<i>4.00</i>
48 h	9.54	4.94	4.18	4.80	<i>2.07</i>
72 h	14.70	7.31	7.31	8.24	<i>5.50</i>
168 h	10.68	8.05	9.86	10.02	<i>6.98</i>
336 h	21.79	11.50	13.07	12.84	<i>8.78</i>
720 h	15.79	13.62	16.69	15.56	14.49
1080 h	16.13	13.68	11.29	14.63	<i>10.65</i>

Mean absolute error in mdeg, against a signal standard deviation of about 52 mdeg. **Bold:** best method that can actually run on a real gap. *Italic:* the driver filler, which cannot — see Section 5.3.

Filler error against gap length, and the calibrated interval

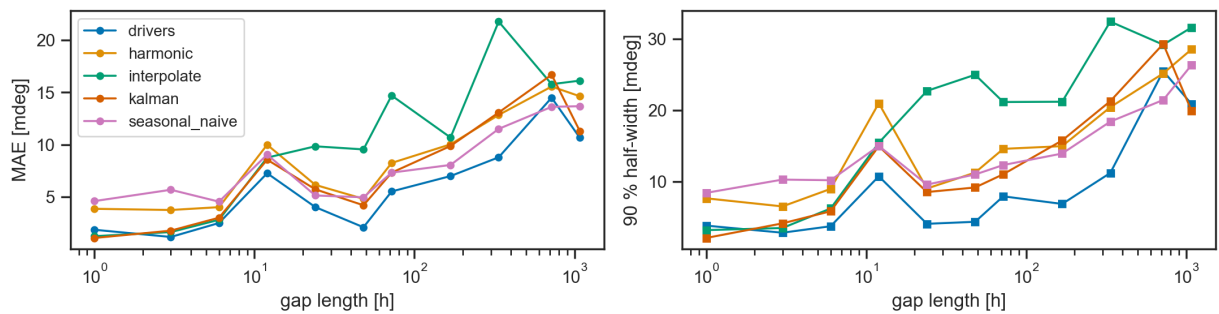


Figure 6: Left: mean absolute error against gap length. Right: the width of the calibrated 90 % interval for the same method and length.

No method wins everywhere, and the crossover is sharp

Below six hours, interpolation and the Kalman smoother are interchangeable and both are excellent — around 1 to 3 mdeg, which is 2 to 6 % of the signal’s standard deviation. Nothing more elaborate helps, because over a few hours the series really is close to a straight line between its endpoints.

Past twelve hours interpolation collapses and never recovers. At 24 hours it is at 9.8 mdeg while the seasonal-naive filler is at 5.1, because a straight line drawn across a day removes the diurnal cycle the data actually contains. This is the single clearest result in the benchmark and it sets the six-hour boundary used in every policy below.

Beyond a day, the choice is between knowing the daily cycle and knowing the local state, and they trade places repeatedly: seasonal-naive wins at 24, 72, 168, 336 and 720 hours, the Kalman smoother at 48 and 1080. The differences are small compared with the error itself, so the policy’s choice among them matters far less than the decision to stop interpolating.

5.2 The uncertainty attached to a filled value

A mean absolute error says how wrong a filler is on average. It does not say how wrong a particular value might be, which is what a consumer needs in order to weight or exclude it. The per-trial errors are therefore used directly as a calibration sample: the interval covering 90 % of them is, by construction, the interval that would have covered 90 % of the errors observed. No distributional assumption is made — these errors are neither Gaussian nor independent, and assuming either would understate them.

Gap	interp.	seas. naive	kalman	harmonic
1 h	3.16	8.39	2.06	7.63
6 h	6.20	10.17	5.81	8.95
24 h	22.71	9.56	8.51	9.00
168 h	21.19	13.92	15.75	14.95
720 h	29.20	21.44	29.31	25.18
1080 h	31.62	26.33	19.94	28.54

Half-width of the 90 % interval, in mdeg.

The interval widens with gap length for every method, as it must. It is also consistently two to three times the mean absolute error, which is the honest consequence of a long-tailed error distribution: the typical filled value is good, and the occasional one is not.

5.3 The driver filler, and why its score is not to be believed

The driver filler has the lowest error at eight of the eleven lengths tested. It is also the one method in the set that **cannot be used at all**.

Section 4 established that every channel of this station fails together: on 100 % of the missing hours, air temperature and humidity are missing too. The benchmark cannot see this, because it injects its gaps where the record is complete — so the drivers are present throughout every injected gap, and absent throughout every real one.

This is the exchangeability assumption, failing measurably

Any conformal guarantee holds under exchangeability between the calibration sample and the deployment conditions. Here the two differ in a way that is not subtle and not statistical: the calibration gaps have their predictors, the deployment gaps do not.

Had the policy been allowed to select on benchmark score alone, it would have chosen the driver filler at eight lengths, and the resulting fill would have degraded silently to interpolation while carrying the driver filler’s error bars — claiming 2.07 mdeg at 48 hours where interpolation delivers 9.54.

The study therefore measures the method and then refuses it. What that refusal costs is small and is stated: between 0.0 and 2.7 mdeg of accuracy per band, 1.05 mdeg on average.

This is the concrete case for the availability criterion that the companion manuscript’s regressor-diagnosis stage applies. At this site the criterion does not merely reorder the candidates — it removes every one of them.

6 The longest stretch

Every extra hour of continuous record costs invented data. The frontier is the exchange rate, and it is the answer this study exists to give.

6.1 Without filling anything

Eight years of archive contain 108 consecutive days

The longest run in which no hour is missing, over the whole record, runs from **2025-05-05 to 2025-08-21** — **108.2 days**, or 0.30 of an annual cycle. It sits in the current era, which is the shortest and best-covered part of the archive.

That is what the record offers to any method requiring strict continuity, and it is why every earlier study in this project tolerated gaps rather than demanding their absence.

6.2 The frontier

Tolerance	Stretch	Cycles	From	To	Fabricated
— (observed)	108.2 d	0.30	2025-05-05	2025-08-21	0.00 %
2 mdeg	169.5 d	0.46	2023-10-30	2024-04-16	0.61 %
5 mdeg	211.5 d	0.58	2021-08-20	2022-03-19	1.60 %
10 mdeg	617.2 d	1.69	2020-07-31	2022-04-09	3.20 %
15 mdeg	988.9 d	2.71	2023-06-21	2026-03-06	14.52 %
20 mdeg	988.9 d	2.71	2023-06-21	2026-03-06	14.52 %
40 mdeg	988.9 d	2.71	2023-06-21	2026-03-06	14.52 %

Fabricated share is measured *inside* the winning stretch, not across the record.

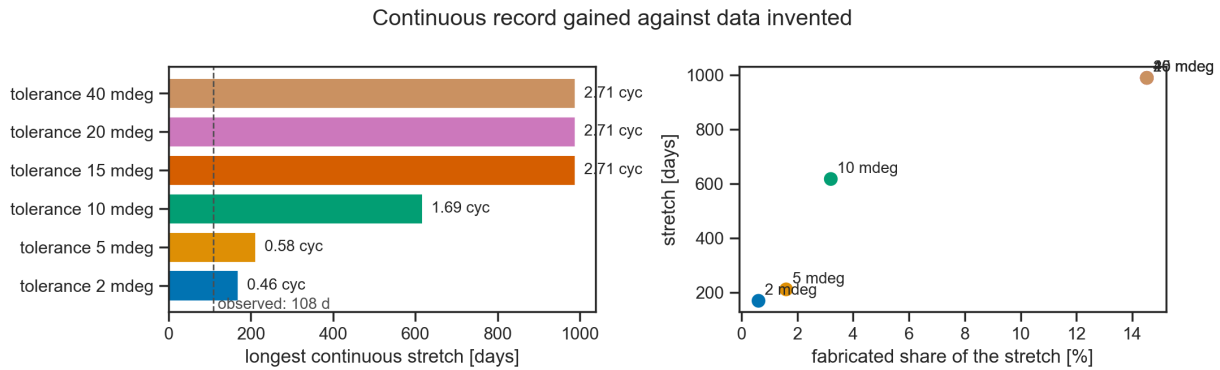


Figure 7: Left: longest continuous stretch per policy, annotated with the annual cycles it contains; the dashed line is what the record offers unfilled. Right: the same stretches against the fabricated share paid for them.

The frontier has one sharp step and then stops

Between 2 and 10 mdeg of tolerance the stretch grows steadily, and everything gained is cheap: at 10 mdeg the record yields **617 days for 3.2 % fabrication**, which is 5.7 times what strict continuity offers.

At 15 mdeg the stretch jumps to **988.9 days**, because that is the point at which the two 41– and 43-day outages of 2024 become fillable and three segments merge into one. The jump costs a step change in fabrication, from 3.2 % to **14.5 %**.

Past 15 mdeg nothing further is bought at any price. The remaining boundaries are the 59-day, 104-day and 438-day outages, all longer than the longest gap the benchmark can calibrate, and the 438-day one is longer than most of the segments it separates. **Raising the tolerance to 40 mdeg — most of the signal’s standard deviation — buys not one additional day.**

6.3 What the era join is worth

The 988.9-day stretch runs from June 2023 to March 2026 and therefore **crosses the February 2025 instrument changeover**. It exists only in the joined column, and so rests on the estimated offset of Section 3.

Confining every stretch to a single era answers what is available without that estimate.

	Stretch	Cycles	Fabricated
Within one era	617.2 d	1.69	3.20 %
Across the join	988.9 d	2.71	14.52 %
Worth of the join	+371.7 d	+1.02	—

The join buys a year, and the gain is not free of doubt

Accepting the estimated offset adds **371.7 days** and takes the record past **2.7 annual cycles** — enough, for the first time anywhere in this project, to identify an annual component from a single continuous window.

It also inherits the unresolved finding of Section 3: the diurnal amplitude differs by 34 % across the changeover, so the two instruments do not share a scale. A stretch spanning them is continuous in level and inhomogeneous in gain.

The recommendation is therefore conditional. For work in the difference domain, or for anything where a change of scale partway through is tolerable, take the 988.9-day stretch. For anything fitting one amplitude across the whole window — a harmonic decomposition, a seasonal model, an amplitude-based diagnostic — take the 617.2-day stretch inside the legacy era, which is homogeneous and costs 3.2 % fabrication rather than 14.5 %.

The recommended stretch: 989 days, 14.5 % imputed

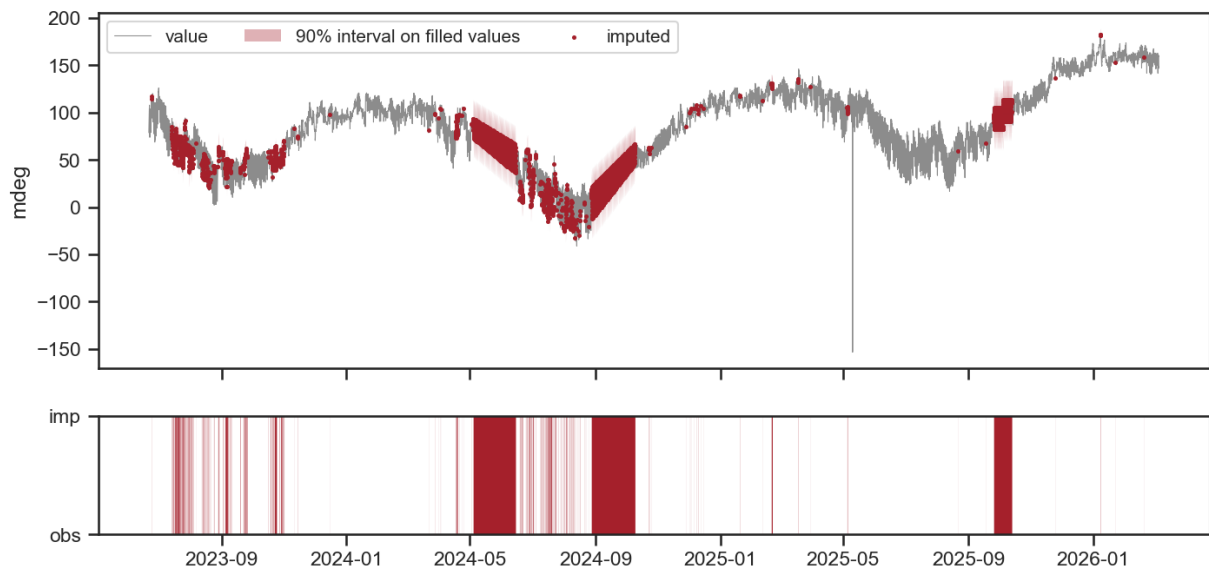


Figure 8: The recommended stretch with its provenance. Upper panel: the series, with imputed values marked and their calibrated 90 % interval shaded. Lower panel: where the imputed values are.

7 Verdicts

Question	Answer
how many stations, and for how long?	st01 1119 days, last reading 2022-06-01; st02 2061 days, still recording; st03 984 days, last reading 2023-06-24. The network failed in stages, not at the changeover.
how complete is the target station?	2061 of 2940 days, 70.1 %. 569 gaps holding 21 098 hours.
where are the missing hours?	84.4 % in six gaps over 30 days. The other 563 gaps together hold 15.6 %, and the 560 shorter than three days hold 8.9 %.
what is the missingness mechanism?	Outage-driven (MNAR). Every covariate is missing on 100 % of the hours the target is missing. No covariate-conditioned method can fill a real gap here.
best filler, short gaps	1 h: kalman, 1.04 mdeg. 6 h: interpolation, 2.79. Below six hours the choice barely matters.
best filler, long gaps	24 h and beyond: seasonal-naive or the Kalman smoother, 4–14 mdeg. Interpolation collapses past 12 h and must not be used there.
the driver filler	Lowest error at eight of eleven lengths, and fail — it cannot run on a real gap. Measured, then refused. Refusing it costs 1.05 mdeg on average.
uncertainty on a filled value	Calibrated without distributional assumption; half-widths of 2–31 mdeg depending on method and length, consistently 2–3× the mean error.
longest stretch, nothing filled	108.2 days (0.30 cycles), 2025-05-05 to 2025-08-21.
longest stretch, after imputation	988.9 days (2.71 cycles), 2023-06-21 to 2026-03-06, 14.5 % invented — but it crosses the instrument changeover.
longest homogeneous stretch	617.2 days (1.69 cycles), 2020-07-31 to 2022-04-09, 3.2 % invented, one instrument throughout.
what still cannot be filled	The 438-day, 104-day and 59-day outages. Raising the tolerance from 15 to 40 mdeg buys not one additional day .

The recommendation

For work in the difference domain, or tolerant of a scale change partway: the 988.9-day stretch, June 2023 to March 2026, 14.5 % imputed, with the per-value intervals attached. It is the only window in this project reaching 2.7 annual cycles.

For anything fitting one amplitude across the window: the 617.2-day stretch, July 2020 to April 2022, 3.2 % imputed, one instrument throughout. It is shorter and it is homogeneous, and for a harmonic or seasonal model that matters more than its length.

In both cases the filled values are marked, and their uncertainty travels with them.

8 Caveats

The compensation coefficient is uncalibrated. Every absolute level in this dataset depends on a value never measured for these instruments. Relative statements are far more robust than levels, and the first difference is invariant to it entirely.

The conformal intervals are calibrated on injected gaps. Their guarantee is marginal and holds under exchangeability between injected and real gaps. Section 4 shows that assumption failing in one identified respect — real gaps have no drivers — so the intervals are a floor on the uncertainty, not a complete account of it.

The joined series reconciles levels, not scales. The 34 % difference in diurnal amplitude across the changeover is unexplained and is not removed by the offset.

One station is analysed. All three are built and classified, and only `st02` is carried through the imputation work, by instruction and for comparability with the two prediction studies.

9 Artefact inventory

Artefact	Content
U_01 – U_03	Station classification, era profiles, and the changeover estimate. Written by the ingest
U_04 – U_07	Classification, era profile, monthly and calendar completeness
U_08 – U_10	Every gap, the length bands, and the nine major outages
U_11, U_12	Missingness mechanism, whole record and current era
U_13 – U_15	Filler benchmark, conformal bands, and the driver illusion
U_15b	What excluding the unavailable filler costs, band by band
U_16	The policies, one per tolerance
U_17, U_18	Stretch frontier: across the era join, and confined within one era
U_19, U_20	The recommended stretch, and the verdicts
U_F01, U_F02	Raw series per station; the era join
U_F03 – U_F05	Station completeness, channel availability, calendar
U_F06	Gap anatomy
U_F07	Filler benchmark and calibrated intervals
U_F08, U_F09	Stretch frontier; the recommended stretch

Reproduction, from the study directory with `neuralprophet_env` active:

```
jupyter --to ipynb build_unified_dataset.py
jupyter nbconvert --to notebook --execute --inplace
    build_unified_dataset.ipynb

jupyter --to ipynb imputation_study.py
jupyter nbconvert --to notebook --execute --inplace
    imputation_study.ipynb
```

The ingest must run first: the study reads its output and nothing else.